

ECF	Error carried forward	SF	Significant figures error	M0	No A marks awarded
AE	Arithmetic error	BOD	Benefit of doubt	^	More is needed in answer
POT	Power of ten error	CON	Contradictory response	XP	Wrong physics
TE	Transcription error	IR	Irrelevant (part) response		

1a	<u>Gradient</u> is not constant so <u>acceleration</u> changes with speed Hence resultant force changes with speed Resultant force consist of weight and air resistance and weight is constant (hence can be deduced that air resistance varies with speed).	A	M1 A1 A0
1bi	$F_{\text{net}} = mg - kv^2$ Using N2L, $F_{\text{net}} = ma$ $mg - kv^2 = ma$ $m(g - a) = kv^2$ $(g - a) = \frac{kv^2}{m}$	A	B1 B1 A0
1bii	At $v = 4.0 \text{ m s}^{-1}$ $\text{Gradient} = \frac{6.0 - 2.8}{0.68 - 0.28} = \frac{3.2}{0.40} = 8.0$ $a = 8.0 \text{ m s}^{-2}$ $g - a = 9.81 - 8.0 = 1.8 \text{ (or 1.81)}$	A	M1 A1 A0
1biii	From b(i), $(g - a) = \frac{kv^2}{m}$, $\frac{(g - a)}{v^2} = \frac{k}{m}$ should be a constant for 4.0 m s^{-1}, $\frac{(g - a)}{v^2} = \frac{1.8}{4.0^2} = 0.11$ or 8.0 m s^{-1}, $\frac{(g - a)}{v^2} = \frac{9.8}{8.0^2} = 0.15$ Hence, values of $\frac{(g - a)}{v^2}$ are not the same, suggestion is incorrect.	A	M1 A1